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High capacity monoclinic Nb_2O_5 and semiconducting NbO_2 composite as high-power anode material for Li-Ion batteries

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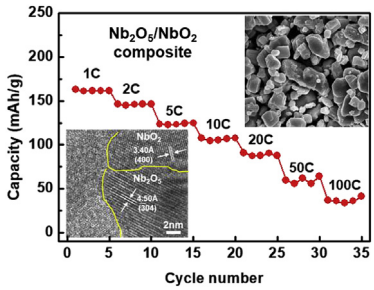
HIGHLIGHTS

- Monoclinic Nb_2O_5 was prepared by high calcination at 1100 °C.
- Semi-conducting NbO_2 was introduced to Nb_2O_5 by thermal reduction process.
- Nb_2O_5/NbO_2 composite shows a cycle performance of 81% retention over 900 cycles.
- Nb_2O_5/NbO_2 composite shows a superior rate capability at 100C.

GRAPHICAL ABSTRACT



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ABSTRACT

Niobium pentoxide, Nb_2O_5, is an intercalation-type material with a high theoretical capacity of ~404 mAh g^{-1} for Li-ion batteries. However, electrochemical properties of Nb_2O_5 largely depend on its various polymorphs with different crystal structures, and their low electrical conductivity acts as the main obstacle. Here, we report high-temperature calcined monoclinic Nb_2O_5 and semiconducting NbO_2 composite as a high-power anode material. Monoclinic Nb_2O_5 itself as a main active material shows a high capacity of ~280 mAh g^{-1}, and NbO_2 with a small band gap of ~0.5 eV not only improves electrical conductivity but also gives a capacity of ~110 mAh g^{-1}. To have a synergic effect of these two materials, the Nb_2O_5/NbO_2 composite is prepared *via* simple post-calcination of as-prepared Nb_2O_5 under a reduction atmosphere. It shows a discharge capacity of ~214 mAh g^{-1} at 0.05 C, a high initial Coulombic efficiency of 94.7%, a superior rate capability of ~40 mAh g^{-1} at 100 C, and a robust cycle performance of 81% retention over 900 cycles.

1. Introduction

Titanium oxide-based materials such as anatase TiO_2 and spinel $Li_4Ti_5O_{12}$ have been extensively explored as promising anode materials owing to their high-power density and safety [1,2]. Their high working voltage window of 1.0–3.0 V *vs.* Li/Li^+ could prevent a significant formation of a solid-electrolyte interphase (SEI) layer and a lithium dendrite, which makes them suitable for microelectronic devices [3]. However, their practical capacities of ~150 mAh g^{-1} for TiO_2 and ~170 mAh g^{-1} for $Li_4Ti_5O_{12}$ are much lower than that of graphite (~350 mAh g^{-1}) [4–7]. Recently, titanium and niobium oxide-based materials such as $TiNb_2O_7$ and Nb_2O_5 have been studied as alternatives owing to their appealing lithium storage properties [8–11]. In comparison to TiO_2 and $Li_4Ti_5O_{12}$ with one possible redox couple of $Ti^{4+/5+}$, those materials have higher theoretical capacities due to multiple redox couples of $Nb^{3+/4+}$ and $Nb^{4+/5+}$ [8]. Among them, niobium

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pentoxide (Nb_2O_5) has attracted attention because of a highest theoretical capacity of ~404 mAh g^{-1}, corresponding to 4 lithium insertion into the unit cell (LixNb_2O_5, $x_{max} = 4$). Research on controlling dimensions of Nb_2O_5 such as spheres, fibers, and nanosheets has been reported to improve the lithium storage properties [12–14]. Based on previous studies [14,15], however, lithium storage properties of Nb_2O_5 more strongly depends on its polymorphs with different crystal structures and lattice parameters; pseudo-hexagonal phase (TT–Nb_2O_5, *P6/mmm*, $a = 3.600$ Å, $c = 3.919$ Å, T = ~600 °C) [16], orthorhombic phase (T–Nb_2O_5, *Pbam*, $a = 6.144$ Å, $b = 29.194$ Å, $c = 3.940$ Å, T = ~800 °C) [17], monoclinic phase (M − Nb_2O_5, *C2/m*, $a = 28.51$ Å, $b = 3.830$ Å, $c = 17.48$ Å, T = ~1000 °C), and high-temperature monoclinic phase (H–Nb_2O_5, *P2*, $a = 21.16$ Å, $b = 3.822$ Å, $c = 19.35$ Å, T > 1000 °C) [18]. Among them, the H–Nb_2O_5 specifically calcined at 1100 °C showed the best electrochemical performances due to the more favorable and larger interlayer spacing for Li^+ ion intercalation/deintercalation despite micro-sized particles caused by crystal growth during high-temperature calcination [14]. In this respect, it is highly recommended to explore monoclinic Nb_2O_5 as the anode material for enhanced electrochemical performances of Li-ion batteries.

Stoichiometric Nb_2O_5 is an insulator with a wide band gap energy of 3.2–4.0 eV, which cause slow electron transfer in the electrode and eventually deteriorate electrochemical performances. To overcome this problem, a few studies have been performed on composite materials such as Nb_2O_5-carbon nanocomposite [19] and Nb_2O_5/graphene composite [20], which results in improved rate and cycle performances compared to those of pure Nb_2O_5. Unfortunately, the Nb_2O_5 in the composites were pseudo-hexagonal [19] and orthorhombic phases [20], which show the much lower specific capacity of ~190 mAh g^{-1} than that of high-temperature calcined monoclinic Nb_2O_5 (~240 mAh g^{-1}) [14]. It is worth noting that if monoclinic Nb_2O_5 are prepared with electrically conductive materials, superior electrochemical performances can be obtained from monoclinic Nb_2O_5-based composites. One facile method is a reduction of Nb_2O_5 by thermal treatment under reducing atmosphere (gaseous H_2, NH_3, *etc.*). This process can normally cause a formation of oxygen vacancy and a reduction of transition metal's valence states, which leads to an improvement in electrical conductivity. It has been reported that non-stoichiometric $Nb_2O_{4.978}$ has a conductivity of $\sim 3 \times 10^3$ S cm^{-1} much higher than $\sim 3 \times 10^{-6}$ S cm^{-1} of the stoichiometric Nb_2O_5 [21]. In addition, another niobium oxide (NbO_2) is an n-type semiconductor with a small band gap of ~0.5 eV that is easily prepared by thermal reduction from Nb_2O_5 [22]. In this respect, a synthesis of non-stoichiometric Nb_2O_{5-x} and Nb_2O_5/NbO_2 composites can be facile strategies to enhance not only electrical conductivity but also electrochemical properties.

Here we first report high-capacity monoclinic Nb_2O_5/semiconducting NbO_2 composite as a high-power anode material for LIBs. To obtain homogeneous-sized and shaped Nb_2O_5 particles, pelletizing and ball milling processes are conducted, and a calcination time is controlled. The NbO_2 phase is introduced to original Nb_2O_5 particles *via* simple thermal treatment under mixed H_2/Ar gases tuned by reduction time and temperature, which plays an important role in lithium storage properties of the final products. We analyze as-prepared composite through various tools such as electron microscopes, X-ray diffraction, X-ray photoelectron spectrometer, Ultraviolet–visible spectroscopy, *etc.* Finally, we demonstrate a great potential of NbO_2/Nb_2O_5 composite for a high-power anode material through half-cell test.

2. Experimental

2.1. Synthesis of Nb_2O_5 polymorphs

As a starting material, Nb_2O_5 particles was purchased (Niobium (V) oxide, 99.9% (metal basis), Alfa Aesar). An XRD measurement revealed that the Nb_2O_5 powders are a composite of pseudo-hexagonal Nb_2O_5 (International Center for Diffraction Data (ICDD), #00-028-0317) and monoclinic Nb_2O_5 (ICDD, #00-027-1312) (Fig. S1). To obtain high-temperature monoclinic phase Nb_2O_5 (H–Nb_2O_5), the Nb_2O_5 powders was annealed at a temperature at 1100 °C in air for different times from 1 h to 10 h.

2.2. Synthesis of NbO_2/Nb_2O_5 composite

To synthesize electrically conductive NbO_2 phase, as-prepared H–Nb_2O_5 powders were annealed under reducing atmosphere. In details, to find out the starting temperature of a formation of NbO_2 phase, the Nb_2O_5 particles were put in an alumina crucible, and then it was placed in the center of a horizontal tube furnace using a quartz tube. The samples were reduced at the temperature of 800 °C for 0 min (ramping rate of 10 °C min^{-1}) under mixed gases, hydrogen (H_2)/argon (Ar) (100/100 sccm). Finally, the samples cooled down to room temperature and were taken out from the furnace.

2.3. Material characterization

As-prepared samples were characterized using various tools as follows; Electrical microscope images were observed using a field emission scanning electron microscope (FE-SEM, JEOL JSM07600F), field emission transmission electron microscope (FE-TEM, JEOL JEM-2100F), and high-resolution transmission electron microscope (HRTEM, JEOL JEM-2100F). X-ray diffraction patterns of samples were obtained by X-ray diffraction analyzer (XRD, Rigaku D/MAX RINT-2000). Rietveld refinement was performed using X'Pert Highscore Plus. X-ray photoelectron spectrometer (XPS, VG Microtech ESCA2000) was used for the analysis of compositional changes. UV-VIS Impedance spectra were gained using impedance analyzer (PARSTAT 2273, Princeton Applied Research).

2.4. Evaluation of electrochemical performance

Coin-type half cells were fabricated to evaluate the electrochemical performance. Working electrodes were prepared using slurries consisting of Nb_2O_5 particles, Nb_2O_5/NbO_2 composite, and the NbO_2 particles as active materials, carbon black powders as a conducting agent and polyvinylidene fluoride (PVDF) as a binder with a weight ratio of 70:20:10. The mixtures were thoroughly mixed with Thinky mixer at a high rotation speed of 2000 rpm. And then, as-prepared slurries were cast on Cu foil with a thickness of ~18 μm, followed by drying in an oven at 100 °C for 30 min. All the working electrodes were dried in a vacuum oven at 120 °C for 5 h before cell assembly. Coin-type half cells (2032R type) were fabricated using working electrodes, a pure lithium metal foil as a counter electrode, 1 M $LiPF_6$ in ethylene carbonate/diethylene carbonate (EC/DEC, 1:1 vol %, PANAX StarLyte, S. Korea) as an electrolyte and a polypropylene (PP) separator. All the coin cells were tested using a battery cycle tester (TOSCAT 3000, Toyo System, Tokyo, Japan).

3. Results and discussion

Bare Nb_2O_5 particles with mixed phases of TT–Nb_2O_5 and M − Nb_2O_5 show an irregular particle shape and a size (Fig. 1a). After calcination at ~1100 °C for 5 h, H–Nb_2O_5 particles were agglomerated that leads to a large and inhomogeneous particle size in the range of 1–5 μm (Fig. 1b). To obtain homogeneous H–Nb_2O_5 particles, pelletizing and ball milling processes were conducted for crushing and grinding. Fig. 1 shows SEM images of pelletized and ball-milled bare Nb_2O_5 particles (Fig. 1c) and H–Nb_2O_5 particles (Fig. 1d). Even after post-annealing at 1100 °C for 5 h, the H–Nb_2O_5 shows a uniform particle size of ~500 nm width and ~2 μm length and a prismatic morphology. The calcination time was also monitored (Fig. S2). The time was reduced to 1 h, the particle size was in the range of 1–2 μm smaller

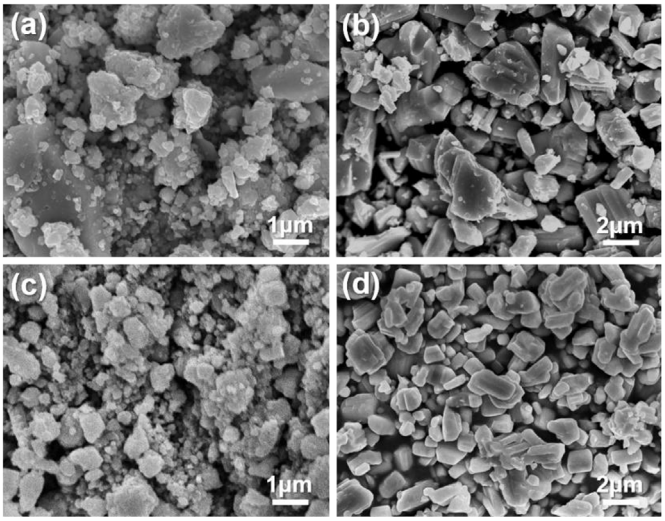

Fig. 1. SEM images of (a) purchased Nb_2O_5 particles before calcination, (b) purchased Nb_2O_5 particles after 1100 °C calcination, (c) pelletized and ball-milled Nb_2O_5 particles before calcination, and (d) pelletized and ball-milled Nb_2O_5 particles after 1100 °C calcination.

than the size of 5 h annealing sample. In contrast, the time was increased to 10 h, severe crystal growth and agglomeration occurred, resulting in a large particle size over 4 μm.

Fig. 2 shows unit cell structures of H–Nb_2O_5 and NbO_2. For the H–Nb_2O_5, it has a layered structure consisting of NbO_6 octahedra ribbons that provide large vacancy and 2D channels through *b* and *c* axes for Li^+ ions. For tetragonal NbO_2 (Space group I41/a, $a = b = 13.710$ Å, $c = 5.985$ Å, $\alpha = \beta = \gamma = 90°$, $V = 1124.97$ Å^3), it has a rutile structure that consists of NbO_x polyhedra lined to form strings. Along the strings, Nb–Nb distances are in the range of 2.8–3.4 Å that create an empty 1D channel along *c* axis (Fig. 2b). It is assumed that the semiconducting NbO_2 improve an electron transfer of overall electrical conductivity of the H–Nb_2O_5 and acts as Li^+ intercalation/deintercalation material through the voids. Moreover, it is known that the nanosized NbO_2 exhibits a pseudo-capacitive property like that of T–Nb_2O_5 (orthorhombic) that could be beneficial for the high-power density anode material [23,24].

To synthesize the NbO_2 phase, the H–Nb_2O_5 particles were annealed at different temperatures and times under the mixed gases of H_2/Ar. The reduction temperatures varied in the range of 600–1000 °C without holding (Fig. S3a). XRD peak patterns indicate that the reduction of H–Nb_2O_5 particles did not occur at 600 °C. The XRD patterns were still matched with the reference of H–Nb_2O_5 phase (ICDD, monoclinic Nb_2O_5 #00-037-1468). When the temperature increased to 800 °C, they start to be reduced to NbO_2 phase that leads to a composite of H–Nb_2O_5 and NbO_2 confirmed by the patterns with two references of Nb_2O_5 and NbO_2 (ICDD, NbO_2 #00-043-1043). As the temperature was set at 1000 °C, the H–Nb_2O_5 phase was transformed to the NbO_2 phase. Reducing times were also studied in the range of 0–60 min at the same temperature of 800 °C to control the ratio of H–Nb_2O_5 and NbO_2 (Fig. S3b). As the reduction time increased, the intensities of NbO_2 phase became strong. A Rietveld refinement method was used for a calculation of a percentage of two phases. H–Nb_2O_5 to NbO_2 are 70.1%/29.9% for 0 min, 23.4%/76.6% for 30 min, and 1.9%/98.1% for 60 min, respectively. This result revealed that the H–Nb_2O_5 phase drastically turned into the NbO_2 phase within 30 min. Fig. 3a–c shows XRD patterns of bare H–Nb_2O_5 particles, H–Nb_2O_5/NbO_2 composite (reduction for 0min), and NbO_2 particles (reduction for 60 min), and their color changes. Fig. 3d exhibits the color change of Nb_2O_5 particles from white to black after the thermal reduction. As a result, three samples were chosen for further characterizations and electrochemical performances to investigate the effect of introducing electrical conductive NbO_2 phase.

A morphological change of three samples by thermal reduction was monitored by SEM in Fig. 4a–c. No significant changes in sizes between samples were observed, but the prismatic structure of bare H–Nb_2O_5 particles became irregular and crooked after reduction. High-magnification SEM images clearly show a change of a surface roughness (Fig. S4). The more particles are reduced, the rougher their surface become due to recrystallization and phase transformation, which is also observed in low-magnification TEM images (Fig. 4d–f). A high-resolution (HR) TEM image of H–Nb_2O_5 with a lattice fringe of ~3.48 Å corresponding to a (402) plane as shown in Fig. 4g. In contrast, an HR-TEM image of H–Nb_2O_5/NbO_2 composite in Fig. 4h shows three distinguished amorphous and crystalline regions such as lattice fringes of

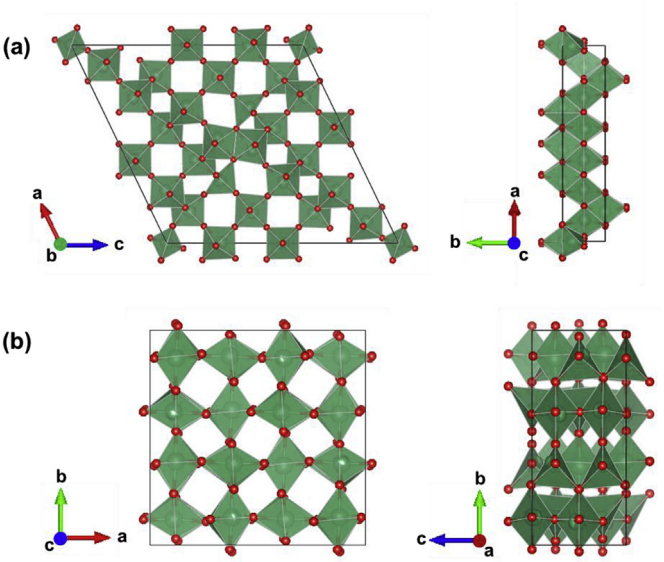

Fig. 2. Illustration of unit cell structures of (a) monoclinic Nb_2O_5 projected along *b*-axis (left) and *c*-axis (right) and tetragonal NbO_2 projected along *c*-axis (left) and *a*-axis (right). Red spheres are oxygen, and the niobium is located at the center of the green coloured polyhedras. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

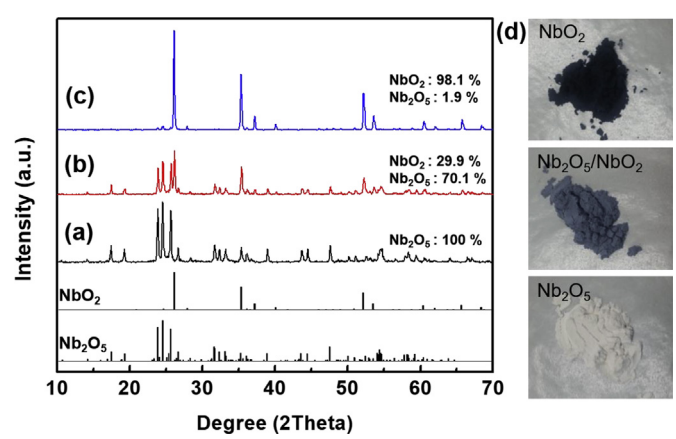

Fig. 3. XRD patterns of (a) H–Nb_2O_5 calcined at 1100 °C, (b) H–Nb_2O_5/NbO_2 composite after reduction at 800 °C for 0 min, and (c) NbO_2 after reduction at 800 °C for 60 min. References of H–Nb_2O_5 (ICDD, monoclinic Nb_2O_5 #00-037-1468) and NbO_2 (ICDD, NbO_2 #00-043-1043) were indexed at the bottom for matching. (d) Photos on the right-side show color changes from H–Nb_2O_5 to H–Nb_2O_5/NbO_2 to NbO_2 through reduction processes. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

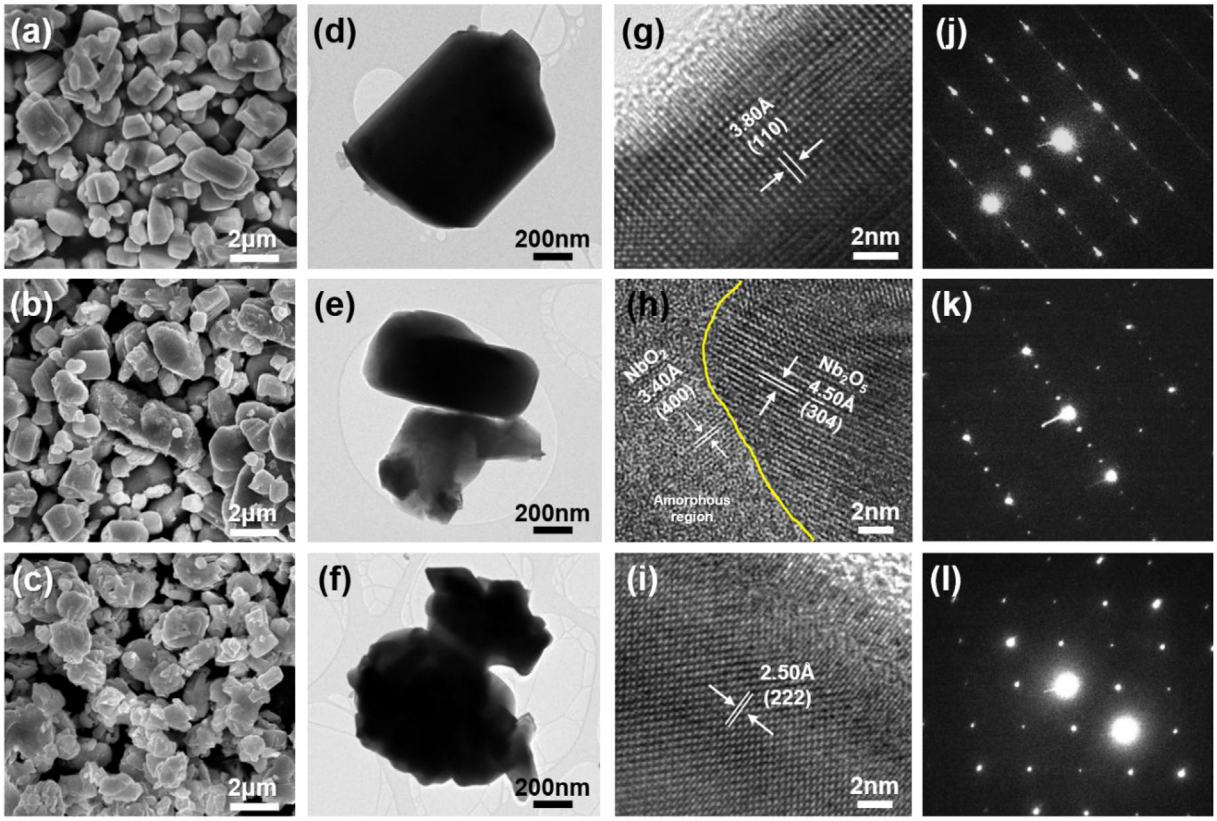

Fig. 4. Morphological and microstructural characterizations of H–Nb_2O_5, H–Nb_2O_5/NbO_2 composite, and NbO_2. (a–c) SEM images, (d–f) low-magnification TEM images, (g–i) HR-TEM images, and (j–l) SAED patterns.

~4.50 Å and 3.40 Å that belongs to ($\bar{3}04$) of H–Nb_2O_5 and (400) of NbO_2, respectively. It is hard to distinguish the lattice constant of the (400) plane of NbO_2 from the lattice fringe of (402) plane of H–Nb_2O_5 due to their very similar d-spacing values. Based on the TEM images, the (400) plane of NbO_2 with 3.40 Å d-spacing was observed particularly near and/or in the amorphous region that newly evolved because of H_2/Ar reduction. Moreover, a low-magnification TEM image reveals that the NbO_2 phase exists as a form of islands in the particles (Fig. S5). Fig. 4i shows a lattice fringe of ~4.50 Å that belongs to a (222) plane of NbO_2. It is worth noting that after full reduction, the sample has high crystallinity without noticeable amorphous regions. Fig. 4j–l confirm that the polycrystalline nature of the samples in agreement with the XRD results.

Fig. 5 shows Nb 3d XPS spectra obtained from the three samples to investigate the valence states. The H–Nb_2O_5 (black line) exhibits doublets of the $3d_{3/2}$ and $3d_{5/2}$ at ~210 eV and 207.3 eV, respectively. This agrees well with the previous report on Nb_2O_5 calcined at a high-temperature at 1050 °C and indicates a presence of Nb^{5+} oxidation state [25]. The H–Nb_2O_5/NbO_2 composite (blue line) shows the same doublets with those of the H–Nb_2O_5 but the higher intensity of the $3d_{5/2}$ peak that is attributed to an evolution of new NbO_2 phase (red sublines) and the coexistence of Nb_2O_5 (black sublines). An area ratio of H–Nb_2O_5 to NbO_2 was determined to be 0.69 to 0.31 by peak splitting, which corresponds to the XRD result. The NbO_2, fully reduced one (red line), has also two peaks at 208.4 and 205.9 eV that well agree with the previous report on NbO_2 and confirm a presence of +4 oxidation state [25]. The XPS result demonstrates peaks shifts to the lower angles and the transformation of the valence state from Nb^{5+} to Nb^{4+} as a degree of the reduction increased [26,27]. It is worth noting that the valence transformation of niobium can result in a formation of new orbital close to the valence band and a decrease in a band gap (Eg) [28], which accordingly improves the surface electronic conductivity of niobium oxide composites and electrochemical performances that makes it suitable for the high-power anode material.

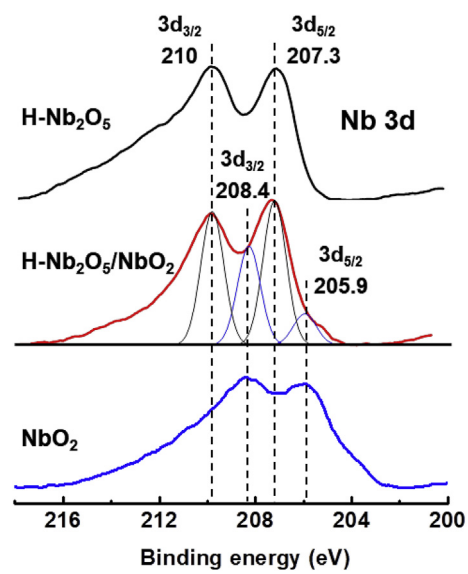

Fig. 5. Nb 3d XPS spectra of H–Nb_2O_5 (black line), H–Nb_2O_5/NbO_2 composite (red line), and NbO_2 (blue line). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

To give a further insight into the changes in colors band gaps of the samples, UV–vis spectroscopy was used. Fig. 6a shows the UV–vis absorption spectra of H–Nb_2O_5, H–Nb_2O_5/NbO_2, and NbO_2. No noticeable intensity was observed from the pristine sample in the visible light range (450 nm ≤ λ ≤ 700 nm) owing to the wide band gap of Nb_2O_5. In contrast, as the degree of the reduction increased, the stronger absorption responses were detected for H–Nb_2O_5 and NbO_2 samples. This

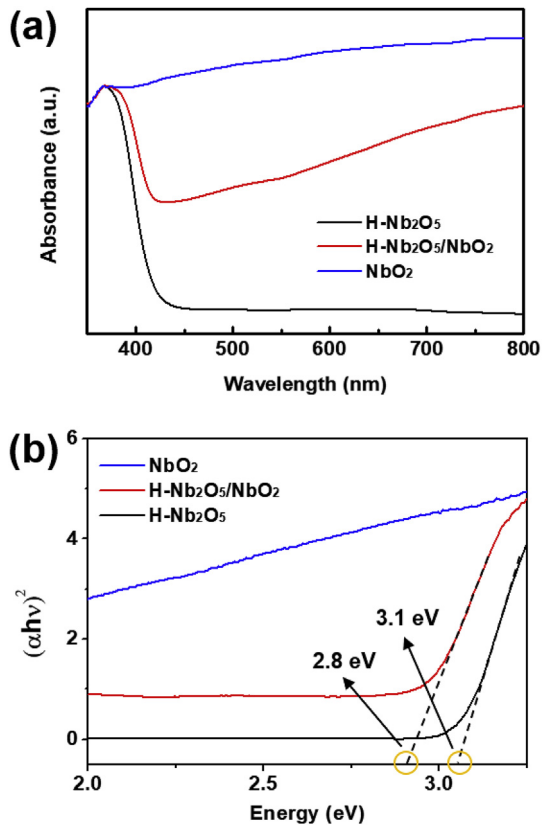

Fig. 6. (a) UV–vis absorption spectra and (b) Tauc's plot of H–Nb_2O_5 (black line), H–Nb_2O_5/NbO_2 composite (red line), and NbO_2 (blue line). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

supports that the strong absorbance in the visible area is attributed to the formation of Nb^{4+} ions and the new energy level lowered than that of the conduction band of Nb_2O_5, which is agreement with the color change of the samples shown in Fig. 2b. Band gaps were calculated from the spectra. According to a Tauc's plot in Fig. 6b, the values of H–Nb_2O_5 and H–Nb_2O_5/NbO_2 composite are 3.1 eV and 2.8 eV, respectively, which indicates the band gap was decreased after reduction.

Electrochemical performances of three electrodes prepared with H–Nb_2O_5, H–Nb_2O_5/NbO_2, and NbO_2 particles as the active materials were evaluated in the in the voltage window of 1.0–3.0 V vs. Li/Li^+ at room temperature. Fig. 7a shows voltage profiles of three samples at a current density of 0.05. They gave first charge (lithiation) and discharge (delithiation) capacities of 283/266, 226/214, and 115/110 mAh g^{-1}, and show high Coulombic efficiencies of 93.9%, 94.7%, and 95.7%, respectively. The reason for the decrease in the capacities of NbO_2/H–Nb_2O_5 and NbO_2 electrodes is attributed to a lower capacity of NbO_2. The oxidation state of Nb in NbO_2 is +3/+4, and it leads to possibly one electron transfer during charging/discharging. Based on our calculation of a theoretical capacity using equation (1), the determined value is ~214 mAh g^{-1} that is almost half of the theoretical capacity of Nb_2O_5 (~404 mAh g^{-1}).

$$Q_{theoretical} = \frac{nF}{3.6 \times M_W} \text{ (mAh g}^{-1}\text{)} \quad (1)$$

Where n is the number of electron transfer per formula unit of the electrode material, F is the Faraday constant, and Mw is the molecular weight of the electrode material. In this work, n is 1 (e^-), Mw is 124.91 g/mol, and F is 96485 sA/mol.

However, it is worth noting that the specific capacity of H–Nb_2O_5/NbO_2 electrode obtained at 0.1 C was still much higher than those of Nb_2O_5 polymorphs including amorphous Nb_2O_5 [15], TT–Nb_2O_5 [14,15,19], T–Nb_2O_5 [14,15,20], and M − Nb_2O_5 [14]. A cycling performance and coulombic efficiency were monitored over 900 cycles in Fig. 7b. Capacity retentions of three electrodes at a high current density of 1 C were 62% (111 mAh g^{-1}), 81% (123 mAh g^{-1}), and 85% (34 mAh g^{-1}), respectively. By introducing NbO_2 to Nb_2O_5, the cycle life was much improved than that of bare Nb_2O_5. Voltage profiles of three samples were plotted in terms of cycles (Fig. S6). Except for a shift to the left due to the capacity fading, no noticeable changes in the curve shapes were observed among samples, which supports that there are no such a decomposition and/or cell failure during cycling similar to those of previous reports on niobium oxide-based materials [29–33]. A reason for the more capacity drop of the H–Nb_2O_5 than others might be attributed to larger intercalation-induced stress/strain to accommodate Li^+ ions into the unit cell [34]. A significant improvement was found in rate capability in Fig. 7c. Although the H–Nb_2O_5 electrode shows a larger capacity at 1C than the others, the H–Nb_2O_5/NbO_2 electrode shows much better property and higher capacity retentions at high C rates from 5 to 100 C. The NbO_2 electrode also shows the superior rate capability, but its capacity is too low. Particularly, the capacity retentions of three electrodes at 100 C were 0.5%, 25%, and 13%, respectively. EIS spectra were obtained to demonstrate a change in a resistance of the samples. Fig. 7d clearly shows the trend of a decrease in the semicircles after reduction, which supports the improved charge transfer of H–Nb_2O_5/NbO_2 and NbO_2 compared to H–Nb_2O_5. It is worth noting that the overall electrochemical performances of H–Nb_2O_5/NbO_2 are superior to those of previous reports, especially on other Nb_2O_5 polymorphs. This is mainly because it has not only enhanced electrical conductivity but also much higher capacity than those of others (amorphous Nb_2O_5: 150 mAh g^{-1}, TT–Nb_2O_5: 180 mAh g^{-1}, T–Nb_2O_5: 140 mAh g^{-1}, M–Nb_2O_5: 200 mAh g^{-1}), which leads to better capacity retentions of H–Nb_2O_5/NbO_2 at even higher current densities [14,15,19,20].

4. Conclusion

In summary, to overcome poor electronic conductivity of high temperature calcined monoclinic Nb_2O_5 with the highest capacity among its polymorphs, semiconducting NbO_2 was introduced to H–Nb_2O_5 *via* simple post-annealing process under reducing atmosphere. The synthetic conditions such as the calcination/reducing temperatures and times were optimized to obtain uniform-sized and prismatic-shaped H–Nb_2O_5/NbO_2 particles for promising electrochemical performances. The XRD analysis revealed that the two phases of H–Nb_2O_5/NbO_2 coexist, and the weight percent was determined to be 70.1/29.9. The electron microspore images confirm that the sample has ~500 nm width and ~2 μm length and consists of three different phases including amorphous region, H–Nb_2O_5, and NbO_2 as a form of the isolated island. As a result, the H–Nb_2O_5/NbO_2 electrode shows the enhanced electrochemical performances including the high-rate capability of the 25% retention at 100 C and the stable cycle life of the 81% retention at 1 C compared to 0.5% and 62% of bare H–Nb_2O_5 electrode. We believe that our experimental strategy provides a way of improvements in monoclinic Nb_2O_5 based-material by surface modification and hope there will be more works on composites with unique conducting materials such as carbon nanotube, graphene, *etc.* to obtain superior lithium storage properties.

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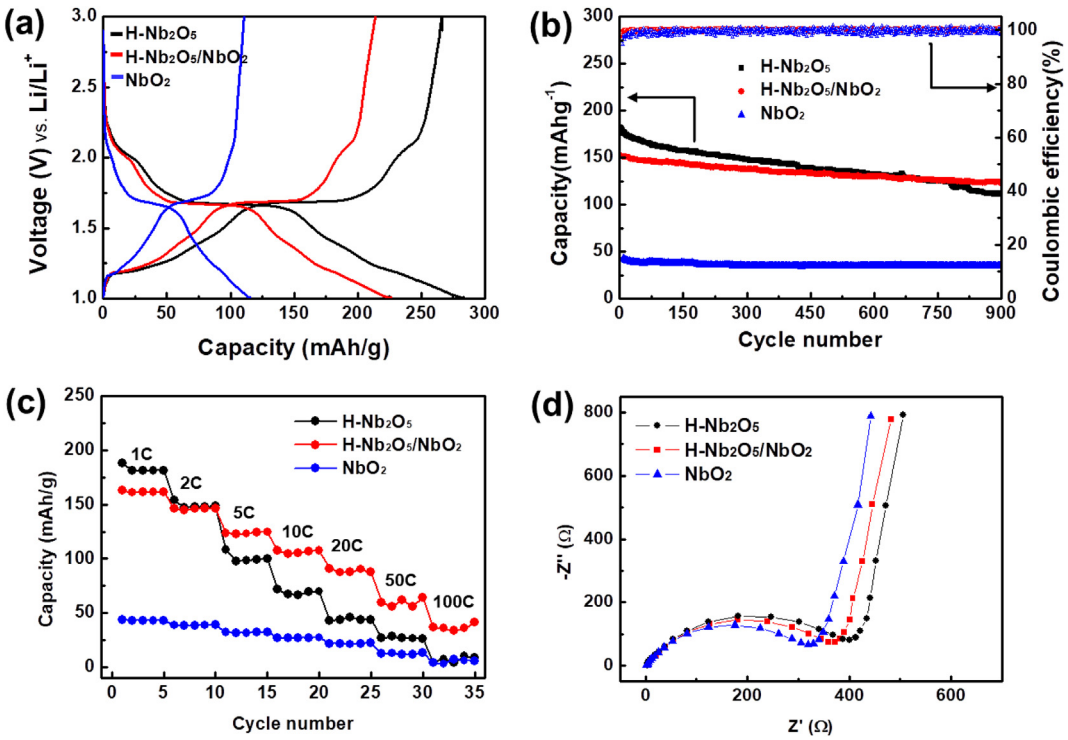

Fig. 7. Electrochemical performances of H–Nb_2O_5 (black line), H–Nb_2O_5/NbO_2 composite (red line), and NbO_2 (blue line) in a voltage window of 1.0–3.0 V vs. Li/Li^+. (a) initial voltage profiles at 0.05C, (b) cycle life at 1C over 900 cycles, (c) rate capability at various current densities from 1C to 100C, and (d) EIS spectra under 10 mV and in the frequency range of 200 kHz to 0.1 Hz. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jpowsour.2019.01.015.

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